Bias and Variance of Estimators

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Introduction

An estimator may differ from the truth for two reasons: systematically (it targets the wrong value on average) and randomly (it bounces around its average). Bias captures the first; variance captures the second. Together they determine how close the estimator is to the truth, as measured by mean squared error (MSE). Many important statistical choices – between raw and shrinkage estimators, between parametric and non-parametric methods, between simple and flexible models – are bias/variance trade-offs.

Prerequisites

The reader should know what an expectation is and should be comfortable simulating repeated sampling in R with `replicate()`.

Theory

For an estimator $\hat{\theta}$ of a population parameter θ:

- **Bias**: $\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta$. An unbiased estimator has zero bias.
- **Variance**: $\text{Var}(\hat{\theta}) = E[(\hat{\theta} - E[\hat{\theta}])^2]$. This is the square of the standard error.
- **Mean squared error**: $\text{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$.

These are linked by the classical decomposition:

$$\text{MSE}(\hat{\theta}) = \text{Bias}(\hat{\theta})^2 + \text{Var}(\hat{\theta}).$$

A biased estimator with very small variance can have lower MSE than an unbiased estimator with large variance. This is the principle behind shrinkage estimators (James-Stein, ridge regression, Bayesian posterior means with informative priors): accept a small bias in exchange for a large variance reduction.

Examples of bias:

- The sample variance with divisor n is biased; with divisor $n-1$ it is unbiased. Both are consistent.
- The MLE of variance in a normal model ($\sigma^2 = \sum(x_i - \bar{x})^2/n$) is biased downward by a factor of $(n-1)/n$.
- The sample standard deviation $s = \sqrt{s^2}$ is biased even though s^2 is unbiased, because the square root is concave.

Assumptions

The decomposition is an identity – it always holds – but its empirical assessment via simulation requires that one can simulate from a known truth. In real data, bias and variance are not directly observable, only MSE-on-a-validation-set or related proxies.

R Implementation

```r
library(ggplot2)
set.seed(2026)

true_sigma2 <- 9
n <- 10
reps <- 20000

mle_var <- function(x) mean((x - mean(x))^2)
ub_var  <- function(x) var(x)

est_mle <- replicate(reps, mle_var(rnorm(n, sd = sqrt(true_sigma2))))
est_ub  <- replicate(reps, ub_var(rnorm(n, sd = sqrt(true_sigma2))))

bias_var <- function(est, truth) {
  c(bias     = mean(est) - truth,
    variance = var(est),
    mse      = mean((est - truth)^2))
}

round(bias_var(est_mle, true_sigma2), 3)
round(bias_var(est_ub,  true_sigma2), 3)
```

Two competing estimators of the population variance are compared: the MLE (divide by n) and the unbiased estimator (divide by $n-1$).

Output & Results

```
    bias variance      mse
   -0.90    14.68    15.49      # MLE, biased downward
    0.01    18.11    18.11      # unbiased estimator
```

For $n = 10$, the MLE is biased (expected value 8.10 vs. truth 9.00) but has lower variance and lower MSE. In this regime, a biased estimator is the better point estimate by the MSE criterion. As n grows, both bias and the gap in MSE vanish.

Interpretation

When reporting an estimator, one should state whether it is unbiased, and whether the alternative (if any) has smaller MSE. For predictive models, the bias/variance decomposition motivates cross-validation: a flexible model has low bias but high variance, so its out-of-sample MSE is minimised at an intermediate complexity chosen by CV.

Practical Tips

- Unbiasedness is a mathematical convenience, not a goal. The MSE criterion generally prefers biased estimators in high-dimensional or low-n settings.
- Bias is reported as "bias $= E[\hat{\theta}] - \theta$"; some authors flip the sign (especially in MCMC diagnostics), so read definitions carefully.
- In regression, predictions at a single input can be decomposed into bias and variance terms; this decomposition motivates regularisation.
- Shrinkage estimators (ridge, Bayesian posterior mean, empirical Bayes) explicitly trade a small bias for a larger variance reduction. This is visible in the MSE comparison.
- When simulating bias/variance to inform methodology choice, use many replicates (10 000+); small-reps estimates of bias are themselves noisy.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Scientific process and research workflow
- R, RStudio, Quarto, and renv toolchain
- Data types, tidy data, accuracy and precision
- Import, joins, and missingness with dplyr